hdf_tools_soundscape: One-page App Summary

Evidence basis: README.md, pyproject.toml, head_hdf_utils.py, psychoacoustics.py, head_hdf_utils_demo.ipynb

What it is

A Python utility toolkit for working with HEAD acoustics .hdf time-data files in soundscape research. It focuses on extracting calibrated stereo audio, computing acoustic metrics, and creating analysis plots.

Who it is for

Primary persona: soundscape or acoustics researchers/analysts using HEAD measurement files and Python workflows.

What it does

- Inspects binary HDF headers with ASCII and HEX previews.
- Parses header metadata: start-of-data, channel count, scan count, delta value, and sampling rate.
- Extracts per-channel calibration values (dB) from header text.
- Reads Left/Right float32 channels; optionally applies calibration to output sound pressure in Pa.
- Exports stereo and mono WAV files via scipy.io.wavfile.
- Computes broadband Leq and short-time RMS dB SPL levels.
- Plots Mark Analyzer-style waveform plus Level-vs-Time charts; includes psychoacoustic wrappers (Zwicker loudness, DIN sharpness, Daniel-Weber roughness) through MoSQITo.

How it works (repo-evidenced architecture)

- Components: head_hdf_utils.py (HDF parsing, calibration, WAV export, SPL metrics, plotting), psychoacoustics.py (MoSQITo wrappers), demo notebook for end-to-end usage.
- Libraries: numpy/scipy for signal I/O and processing; matplotlib for plots; mosqito for psychoacoustic metrics.
- Data flow: .hdf file -> header parse + binary float32 read -> Left/Right arrays -> optional calibration to Pa -> metrics/plots and optional WAV/PNG outputs.
- Services or networked backends: Not found in repo.
- Persistent database/storage layer beyond local files: Not found in repo.

How to run (minimal getting started)

- Install dependencies: run `uv sync` (README evidence).
- Use the demo workflow in `head_hdf_utils_demo.ipynb` and set your HDF path in notebook cells.
- Run notebook cells that call `read_head_file(...)`, `plot_mark_style(...)`, and optional psychoacoustic functions.
- Dedicated CLI entrypoint command: Not found in repo.
- Formal test command documentation: Not found in repo.